Analytic Partial-Residual Targets: A Controlled Mechanism Study and its Structural Boundary

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Draft generated from recorded experiment artifacts

Abstract

We study an exact partial-residual target update for grouped additive linear regression. The update is an instrumented form of block-coordinate minimization/backfitting, rather than a learned target generator. In the registered S1 synthetic additive matrix, it met the two registered endpoints against development-selected Adam on five final seeds, but failed the quality endpoint on a separate longer-horizon family. We then registered two structural-misspecification successors with interleaved supplied groups and unmodeled nonlinear/interacting targets. S2.1 adds a common affine intercept after S2 exposed that the initial nonlinear family had a mean-offset confound. S2.1 failed both registered endpoints on its final and long-horizon families. These internally executed synthetic findings establish a narrow boundary, not general learning superiority.

1 Question and mechanism

For a grouped additive model $f(x) = \sum_g f_g(x_g)$, the proposed update constructs $t_g = y - \sum_{h \neq g} f_h(x_h)$ and solves a ridge subproblem for group g. The method uses support labels and current predictions only. It has no teacher, learned inverse, component labels, or trained target controller. Consequently, its most accurate interpretation is partial-residual backfitting/block-coordinate minimization [4, 1], not a general target-propagation result [3, 2].

2 Protocols and evidence boundary

S1 freezes selection on meta-training and development task families, with final and long-horizon labels excluded from selection. It compares tuned full-batch Adam, sequential analytic residual updates, a no-derivation block-gradient ablation, and fixed normalized LMS under deterministic operation-proxy accounting. S2 and S2.1 use distinct versioned protocols and seeds. S2.1 keeps the same information boundary but adds a common learned intercept to all conditions.

The tables and figures are generated from saved outputs and independently read back; no numeric result is entered by hand. The exact commands, configuration hashes, and manifests are retained alongside each run.

3 Results

In S1, the analytic method passed the five-seed final matrix but the longer-horizon matrix failed the registered quality criterion. Thus even the original additive result does not support a persistent long-budget advantage. S2's structural matrix was completed and verified, but its nonlinear target

Table 1: Recorded equal-resource evaluation NMSE. "Fail" means the predeclared composite endpoint was not supported; it does not mean a runtime error.

Study	Split	Adam	Analytic residual	Registered decision
S1	final	0.024140	0.003475	supported on admitted matrix
S1	long horizon	0.004701	0.004754	fail
S2.1	final	0.724612	0.711624	fail
S2.1	long horizon	0.822250	0.822343	fail

Table 2: S2.1 descriptive paired result based on the five frozen task pairs per split. Intervals are deterministic percentile-bootstrap intervals for the mean paired NMSE difference (analytic residual minus Adam).

Split	analytic wins	mean difference	95% interval
final	3/5	-0.0129873	[-0.0281100, 0.0021354]
long horizon	1/5	0.0000932	[-0.0003178, 0.0004638]

had a nonzero mean while no model fitted an intercept; it is retained as a negative run rather than used as a clean structural attribution. S2.1 corrects that common-capacity omission prospectively. It fails the registered quality and efficiency criteria in both final and long-horizon splits. Its paired bootstrap intervals are descriptive only: the protocol endpoints, not post-hoc intervals, control the claim decision.

4 Limitations and negative evidence

The work remains synthetic, CPU-only, internally executed, and small-seed. It has no independent reproduction, no real-data admission track, no neural block implementation, no learned controller, and no wall-clock or memory comparison under a separately frozen protocol. The simultaneous/Jacobi residual ablation is numerically unstable in S2/S2.1 under the selected settings; this is retained as a result, not suppressed. The paper therefore makes no publication-ready claim of general learning utility, deep credit assignment, or learned-target superiority.

5 Reproducibility

Run `sh scripts/run_s21_reproduction.sh OUTPUT_DIRECTORY` from the repository root, then inspect the generated manifest and evaluation-verification receipts. The existing S1 and S2 artifacts are preserved under `runs/`. The external reproduction packet remains unfinished.

References

[1] Andreas Buja, Trevor Hastie, and Robert Tibshirani. Linear smoothers and additive models. *The Annals of Statistics*, 17(2):453–510, 1989.

[2] Max Jaderberg et al. Decoupled neural interfaces using synthetic gradients. *arXiv:1608.05343*, 2016.

[3] Dong-Hyun Lee, Saizheng Zhang, Asja Fischer, and Yoshua Bengio. Difference target propagation. *ECML PKDD*, 2015.

[4] Stephen J. Wright. Coordinate descent algorithms. *Mathematical Programming*, 151:3–34, 2015.